

Calculus II 第十二章

考点:向量, 直线方程和平面方程

1. Multiple Choice Questions:

(1) (2019年期中) Let $\vec{a}$ and $\vec{b}$ be two nonzero orthogonal vectors, which of the following must be true?

- (A) $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$; (B) $|\vec{a} - \vec{b}| = |\vec{a}| - |\vec{b}|$ (C) $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$; (D) $\vec{a} + \vec{b} = \vec{a} - \vec{b}$

(2) (2019年期中) The equations of two lines are $l_1 : x = t, y = 2t, z = -t$, and $l_2 : x = 1 - 2t, y = t, z = -1 + t$. Then l_1 and l_2 are

- (A) parallel; (B) orthogonal; (C) intersect with each other; (D) skew.

(3) (2021年期中) Identify the surface of $2x^2 + y^2 = z^2$.

- (A) Hyperboloid of two sheets; (B) Elliptical Cone;
(C) Hyperboloid of one sheet; (D) Elliptical paraboloid.

(4) (2023年期中) If P_1 and P_2 are the following planes: $P_1 := \{(x, y, z) : x + 2y - z = \sqrt{2}/2\}$, $P_2 := \{(x, y, z) : x + y = -\sqrt{2}/2\}$, then the angle θ between P_1 and P_2 is:

- (A) $\frac{\pi}{2}$; (B) $\frac{\pi}{3}$; (C) $\frac{\pi}{4}$; (D) $\frac{\pi}{6}$.

2. Determine whether the following statements are true or false? No justification is necessary.

(1) (2020年期末) If $\vec{u} \neq 0$, and if $\vec{u} \times \vec{v} = \vec{u} \times \vec{w}$ and $\vec{u} \cdot \vec{v} = \vec{u} \cdot \vec{w}$, then $\vec{v} = \vec{w}$.

(2) (2019年期末) The plane $x + y - 2z = 1$ is perpendicular to the plane $x + y + z = 1$.

3. Please fill in the blank for the questions below.

(1) (2023年期中) The distance from the point $S(1, 1, 1)$ to the plane $x + 2y + 3z + 4 = 0$ is ____.

(2) (2021年期中) If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors and $\vec{a} + \vec{b} + \vec{c} = 0$, then $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} =$ ____.

(3) (2021年期中) If the vector $\vec{c}$ is perpendicular to $\vec{a} = (1, 2, 1)$ and $\vec{b} = (-1, 1, 1)$ and $\vec{c} \cdot (\vec{i} - 2\vec{j} + \vec{k}) = 16$, then $\vec{c} =$ ____.

(4) (2020年期末) The equation of the plane through the line $x = -1 + 2t, y = 3 + t, z = -t$ and parallel to the line $x = -2t, y = t, z = 1 - t$ is ____.

4. (2023年期中) Let $L_1 : x = t, y = 1, z = 1 + t$ and $L_2 : x = 1 + 2s, y = 0, z = 2s$. Is there one plane in which both lines lie? If so, find the equation of the plane. If not, give your reason.

5. (2022年期末) Find the equation for the plane through the origin parallel to the following lines: $l_1 : x = 1, y = -1 + t, z = 2 + t$, and $l_2 : x = -1 + t, y = -2 + 2t, z = 1 + t$.

6. (2021年期末) Find the equation of the plane through point $(1, 0, 1)$, and perpendicular to the plane $x - 2y + 3z + 2 = 0$ and the plane $x + 2y - 3z - 2 = 0$.

7. (2019年期中) Find the equation of the plane through the points $(2, -1, -1)$ and $(1, 0, -1)$ perpendicular to the plane $2x + 3y - 5z + 6 = 0$.